

Selim Kuzucu

PhD researcher in efficient vision-language models and scalable data pipelines for training and evaluating LVLMs

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EDUCATION

- **Max Planck Institute for Informatics & Google (ELLIS Industry Track)** Saarbrücken, DE
PhD in Computer Science Sept 2024 - Current
- **Middle East Technical University** Ankara, TR
Bachelor of Science - Computer Engineering; GPA: 3.66/4.00, **top 5% of the class** Aug 2019 - June 2023
Activities: Founder of Management Consulting Club, Member of ACM Student Chapter & Robotics Club

SELECTED PUBLICATIONS

- **S. Kuzucu**, A. Tonioni, V. Lup, B. Schiele, F. Tombari, M. F. Naeem, “*PARCEL: Pool-Anchored Resampling with Conditioned Elastic Queries for Efficient Vision-Language Understanding*”. **under review** | arXiv | project page.
Elastic LVLM inference across 30+ benchmarks, 2 LVLM backbones and different visual token budgets.
- M. Farina*, V. Udandarao*, T. Nguyen*, **S. Kuzucu**†, M. Boether†, A. Hochlehnert†, A. Ghosh†, M. Nezhurina†, et al., “*DataComp-VLM: Improved Open Datasets for Vision-Language Models*”. **NeurIPS 2026 E&D** | arXiv | Hugging Face.
Data-centric VLM scaling study over **160 datasets**, **6T multimodal tokens**, **52 evaluations**, large annotation jobs, **3B+ LVLM pretraining**, and **300K+ Hugging Face downloads**.
- **S. Kuzucu**, M. F. Naeem, A. Kukleva, F. Tombari, B. Schiele, “*Enhancing Self-Supervised Visual Representation Learning via Low-Rank Adapted LLMs*”. **Transactions on Machine Learning Research (TMLR) & NeurIPS 2026 VLM4RWD Workshop** | OpenReview.
Improves self-supervised visual representations by adapting frozen LLM blocks with low-rank updates.
- **S. Kuzucu***, K. Oksuz*, J. Sadeghi, P. K. Dokania, “*On Calibration of Object Detectors: Pitfalls, Evaluation and Baselines*”. **ECCV 2024 Oral** | proceedings.
Oral presentation; top 2.3% of valid submissions, establishes stronger baselines for detector calibration.
- K. Oksuz, **S. Kuzucu**, T. Joy, P. K. Dokania, “*MoCaE: Mixture of Calibrated Experts Significantly Improves Object Detection*”. **Transactions on Machine Learning Research (TMLR)** | OpenReview.
Combines calibrated detectors into stronger expert mixtures, improving object detection performance across benchmarks.
- **S. Kuzucu**, J. Cheong, S. Kalkan, H. Gunes, “*Uncertainty-based Fairness Measures*”. **Journal of Artificial Intelligence Research (JAIR)** | paper.
Introduces uncertainty-aware fairness measures that expose subgroup disparities missed by point-estimate metrics.

EXPERIENCE

- **Max Planck Institute for Informatics** September 2024 - Present
Doctoral Researcher **Advisors: Prof. Dr. Bernt Schiele & Dr. Federico Tombari**
 - Co-developed **DataComp-VLM**, a data-centric VLM scaling study spanning **160 public datasets**, **6T multimodal tokens**, **52 evaluations**, **3B+ parameter LVLM pretraining**, and **170K+ Hugging Face downloads**.
 - Scaled data and annotation pipelines for large multimodal corpora, connecting dataset design, synthetic/curated annotations, pretraining, and downstream evaluation.
- **Google Zurich** September 2025 - May 2026
Student Researcher **Supervisors: Dr. Ferjad Naeem & Dr. Federico Tombari**
 - Developed **PARCEL**, an elastic LVLM inference method that improves the performance-efficiency trade-off across **27 benchmarks** and **16–256 visual-token budgets** against existing matryoshka baselines.
 - Ran large-scale **JAX experiments** on distributed TPU infrastructure, regularly using around **512+ TPUv4/TPUv5e chips** for LVLM training and evaluation.
- **Five AI (Bosch UK) - Oxford Applied Research Center** September 2023 - June 2024
Intern Research Scientist **Supervisor: Dr. Puneet K. Dokania**
 - Led **object-detector calibration benchmarking** for ECCV 2024 Oral work, identifying evaluation pitfalls in reliability metrics and calibration protocols.
 - Adapted **post-hoc calibration methods** to object detection, outperforming state-of-the-art train-time calibration techniques by **90%+** on key calibration metrics.
 - Released an **open-source calibration library** for reproducible detector calibration and evaluation, available on **GitHub**.

- **AFAR Lab at University of Cambridge**

February 2022 - August 2023

- Undergraduate Student Researcher

Supervisor: Prof. Dr. Hatice Gunes

- **Introduced uncertainty-based fairness measures** that reveal subgroup bias missed by point-prediction fairness metrics.
- **Validated the proposed fairness gap** through experiments on **three synthetic and three real-world datasets**, showing how uncertainty discrepancies expose hidden model bias.
- **Studied gender fairness in mental-health prediction**, evaluating mitigation strategies across multimodal affective-computing pipelines.

- **METU Image Lab**

October 2021 - September 2022

- Undergraduate Student Researcher

Supervisor: Prof. Sinan Kalkan

- **Developed uncertainty-based methods** for mitigating class imbalance in visual recognition.
- **Curated and benchmarked SVCI'20**, a semantically imbalanced dataset for evaluating modern imbalance-mitigation techniques.

PROJECTS

- **Code Implementation for Uncertainty-Aware Learning Against Label Noise on Imbalanced Datasets (AAAI'22)** Provided the unofficial implementation for all methods in the paper when official code was unavailable. Conducted as part of CENG502: Advanced Deep Learning. Available on [GitHub](#) (June 2023).
- **Code Implementation for Uncertainty Quantification in CNN Through the Bootstrap of Convex Neural Networks (AAAI'21)** Provided the unofficial implementation for all methods in the paper when official code was unavailable. Available on [GitHub](#) (June 2022).

HONORS & AWARDS

- Earned Turkish Presidency Extraordinary Young Talent Award for academic work and extracurricular activities during college and high school - September 2023
- First place in Guided Research Symposium at METU Computer Engineering Department among 30 undergraduate research projects - June 2022
- Finalist, Swarm Robotics Competition at Teknofest 2021 (top 10 out of 150 teams) - August 2021
- Earned METU Development Foundation's Academic Merit Scholarship - September 2019
- Ranked in top 0.01% (270th) in Turkish university entrance exam among 2.5 million test takers - July 2019

SKILLS SUMMARY

- **Languages** Python, C++, JavaScript, C#, C, Bash, MySQL
- **ML Systems** JAX, PyTorch, distributed GPU/TPU training, large-scale pretraining, multimodal evaluation
- **Frameworks and Platforms** MMDetection & MMCV, ROS, Angular, Flask, Linux
- **Interests** Bass and electric guitar, Baduk (Go/Weiqi), Chinese language and literature, gothic and East Asian literature

TEST SCORES

- GRE: 328/340 (Verbal: 160/170, 85%; Quantitative: 168/170, 90%; Analytical Writing: 4.5/6, 79%) - October 2022
- TOEFL: 118/120 (Reading: 30, Listening: 29, Speaking: 30, Writing: 29) - August 2021